

Predicting Whether a Barcelona Bicing Station Will Become Empty Within the Next 60 Minutes

1. Introduction

Bike-sharing systems are a useful part of urban mobility, but their usefulness depends on the availability of bicycles at the right place and time. In this project, the focus is specifically on one operational problem: stations becoming empty and therefore being unable to serve users who want to start a trip.

The main research question was: **Can we predict whether a Barcelona Bicing station will become empty within the next 60 minutes using its current status, recent bike availability history, time-based information, and location?**

The project was designed as a binary classification problem. For every eligible observation, the target was defined as:

- 1: the station becomes empty at least once during the following 60 minutes while it is open.
- 0: the station does not become empty during that period.

Several approaches were compared, ranging from simple baselines to machine learning models and a neural network. The models included a Dummy Classifier, a rule-based baseline, Logistic Regression, a Decision Tree, Random Forest, and a Neural Network.

2. Dataset

The dataset contains historical status snapshots from [Barcelona Bicing stations](#) between August 2018 and March 2019. Each observation describes the status of a station at a particular timestamp rather than an individual bicycle-trip record.

The main variables used in the project were:

- Station ID
- Timestamp
- Number of available bikes
- Number of free slots
- Station status
- Latitude
- Longitude

After loading and initial cleaning, the dataset contained approximately **24.4 million observations**.

Several basic data quality checks were performed. There were no missing values in the variables selected for the project, no invalid timestamps, and no negative values for bikes or slots. Duplicate station-timestamp records were also identified and removed.

A small number of observations represented stations marked as open while both the number of bikes and free slots were zero. These observations represented only around **0.015% of open-station records** and were removed as operational anomalies.

3. Data Coverage and Temporal Limitations

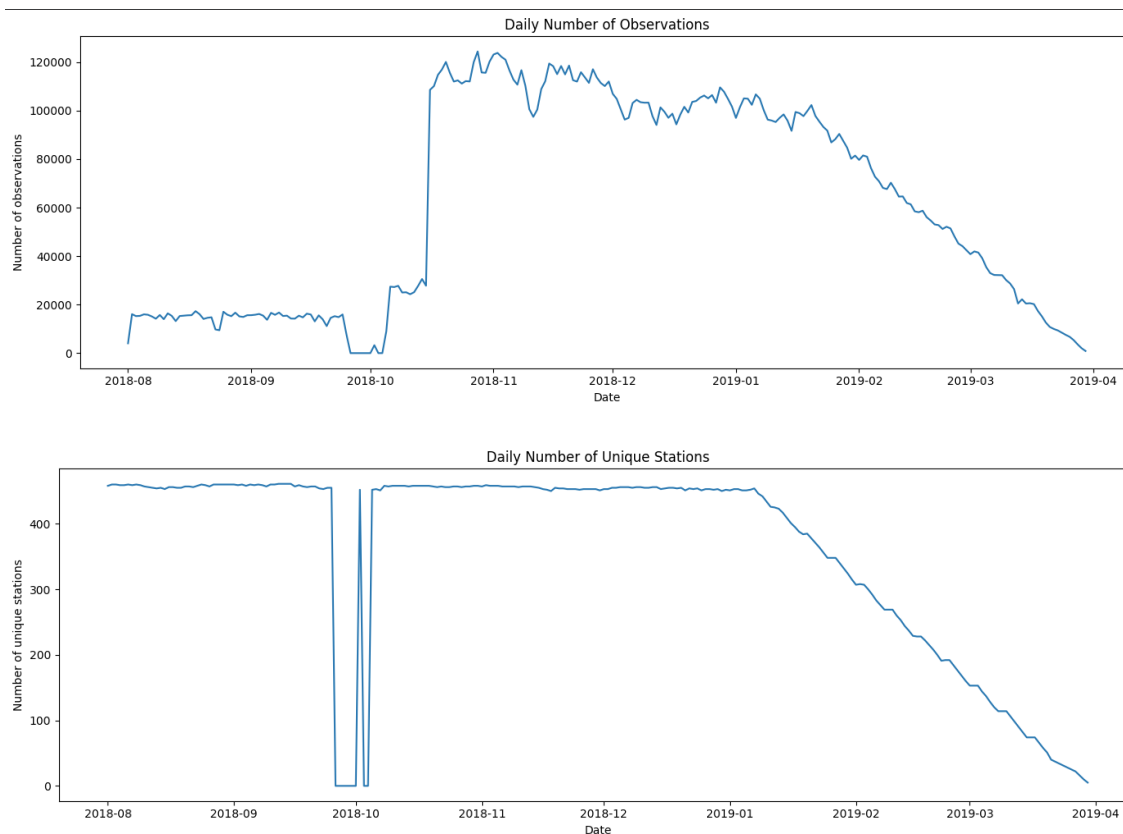
An important part of the project was examining whether data coverage remained stable over time.

Two daily coverage measures were analysed:

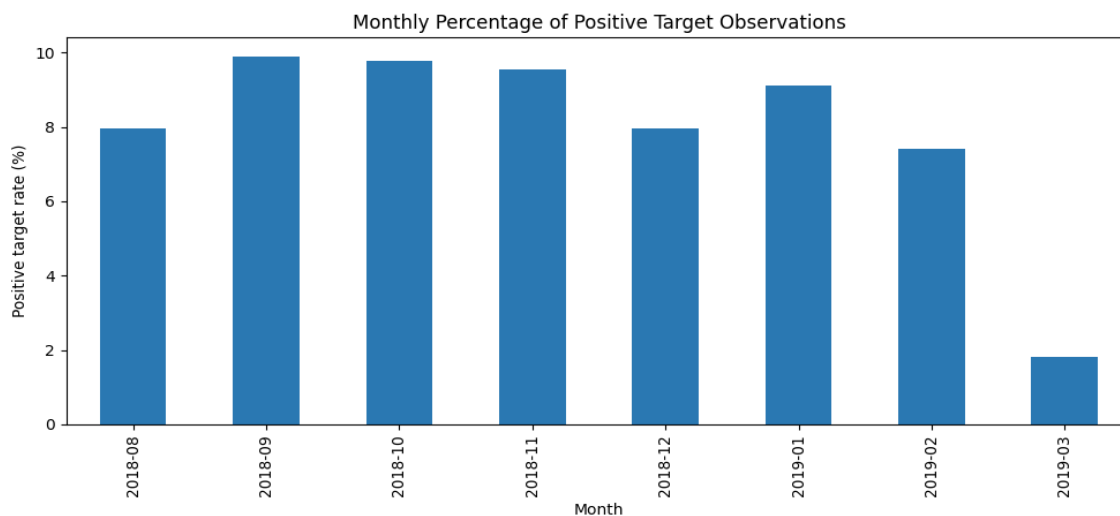
- 1- Number of observations per day
- 2- Number of unique stations represented per day

We had several important issues. There was a clear data gap around late September and early October 2018, when the number of available observations temporarily dropped to zero or near zero.

Station coverage was relatively stable until early January 2019, with approximately 450 stations represented daily. After this point, the number of stations began to decline progressively. The reduction continued throughout February and became particularly severe in March. By the end of March, only a very small number of stations remained in the data.



The monthly positive target rate also showed a substantial change:



The very low positive rate in March coincided with the severe decline in station coverage. For this reason, March was not used in the main model evaluation. Its data was considered insufficiently comparable with the earlier months.

The final modelling period was therefore organised as:

Training set: August to December 2018

Validation set: January 2019

Test set: February 2019

This chronological split preserved the temporal order of the data and avoided training a model using information from future periods.

The final split contained:

Dataset	Observations	Positive rate
Train	9,521,722	9.00%
Validation	2,966,095	9.11%
Test	1,711,745	7.42%

4. Target Definition

Only stations that were currently open and had at least one available bike were considered eligible for prediction. The target was defined as positive when the station reached zero available bikes while still open at least once during the following 60 minutes.

For example, if a station had four available bikes at 12:00 and reached zero bikes at 12:45, the observation at 12:00 received: target = 1

If the station did not reach zero before 13:00, it received: target = 0

5. Feature Engineering

Current station conditions

Number of available bikes

Number of free slots

Observed capacity: sum of available bikes and free slots in each snapshot

Occupancy ratio: number of available bikes relative to observed capacity.

Recent historical availability

To capture recent trends, the number of available bikes approximately 15 and 30 minutes earlier was retrieved for the same station: `bikes_15m_ago/bikes_30m_ago`

Two change variables were then calculated: `bike_change_15m / bike_change_30m`

For example, a negative value for `bike_change_15m` indicates that bikes were removed from the station during the recent period.

Rows without sufficient recent history were removed.

Time based variables: The following temporal features were also included: Hour of the day / Day of the week / Weekend indicator / Month

Location: Latitude and longitude were included to allow the models to capture possible geographical differences between stations.

6. Modelling Strategy

The project compared six different predictive approaches:

Dummy Classifier / Rule-Based Baseline / Logistic Regression
Decision Tree / Random Forest / Neural Network

The models were compared on the validation set using: Accuracy, Precision, Recall

Because approximately 9% of the observations were positive, accuracy alone could be misleading. A model predicting every observation as negative could still obtain an accuracy above 90%.

7. Baseline Models

Dummy Classifier always predicted the majority class. This demonstrates why accuracy alone is insufficient. The model achieved high accuracy simply because most observations belonged to the negative class, but it failed to identify a single future empty station event.

Its validation results were: Accuracy: **90.89%**, Precision: **0%**, Recall: **0%**

Rule-Based Baseline: A simple operational rule was also tested: Predict that a station is at risk when it currently has two or fewer available bikes. The results were: Accuracy: **89.17%**, Precision: **43.32%**, Recall: **61.22%**

This simple rule achieved the highest recall of all tested methods, but its lower precision means that it generated many false alarms.

8. Machine Learning Models

Logistic Regression

Logistic Regression was used as a simple linear machine learning model. A pipeline combined feature scaling with model training, and the scaler was fitted only on the training data. Validation performance: Accuracy: **91.68%**, Precision: **61.31%**, Recall: **23.69%**

Compared with baseline, Logistic Regression improved precision considerably but identified a much smaller percentage of real empty station events.

Decision Tree

A Decision Tree was used to capture nonlinear relationships between current bike availability, recent changes, time, and location.

A limited comparison of different tree depths was performed, and the selected configuration used a maximum depth of 12 and a minimum leaf size of 100. Validation performance: Accuracy: **93.17%**, Precision: **67.50%**, Recall: **48.28%**

The Decision Tree provided a strong balance between precision and recall.

Random Forest

The objective was to examine whether combining multiple trees could improve performance relative to a single Decision Tree.

Validation performance: Accuracy: **93.22%**, Precision: **69.61%**, Recall: **45.53%**

Random Forest achieved the highest accuracy and precision among the main machine learning models, but its recall was slightly lower than that of the Decision Tree.

Neural Network

A Neural Network was also evaluated. The inputs were scaled using a pipeline, and hidden-layer sizes of 16, 32, and 64 neurons were compared on the validation set. The selected configuration used one hidden layer with 16 neurons.

The final validation results were: Accuracy: **92.65%**, Precision: **68.39%**, Recall: **36.00%**

The tuned Neural Network performed substantially better than the initial configurations in detecting positive cases. However, its recall remained lower than that of the Decision Tree, so it was not selected as the final model.

9. Model Comparison and Final Selection

	Model	Accuracy	Precision	Recall
0	Dummy Classifier	0.908863	0.000000	0.000000
1	Rule-Based Baseline	0.891650	0.433186	0.612233
2	Logistic Regression	0.916826	0.613054	0.236899
3	Decision Tree	0.931672	0.674959	0.482750
4	Random Forest	0.932240	0.696076	0.455301
5	Neural Network	0.926509	0.683902	0.360022

Random Forest achieved slightly higher accuracy and precision, but **the Decision Tree identified a larger proportion of the actual empty station events**. Because missing a station at genuine risk is operationally important, the Decision Tree was selected as the final model. It increased recall from 45.53% to 48.28%, while precision decreased only moderately from 69.61% to 67.50%.

For this problem, identifying stations that are genuinely at risk of becoming empty is important. Missing too many positive cases reduces the usefulness of the system.

Although the rule-based baseline achieved higher recall, its substantially lower precision produced considerably more false alarms.

10. Final Test Performance

The selected Decision Tree was evaluated once on the unseen February test set.

The final performance was: **Accuracy: 94.05%, Precision: 63.66% , Recall: 46.23%**

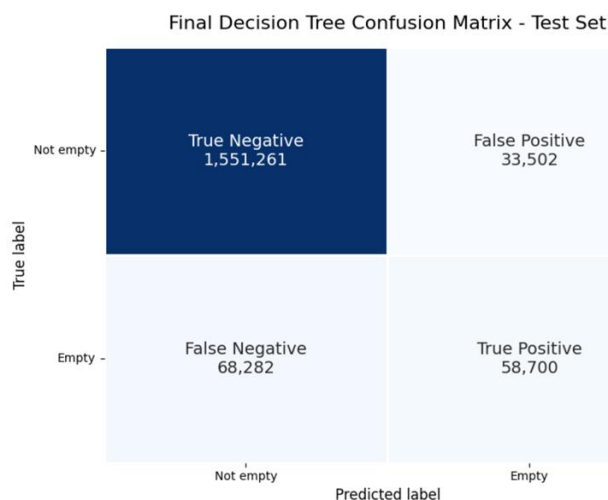
The confusion matrix contained:

True Negatives: **1,551,261**

False Positives: **33,502**

False Negatives: **68,282**

True Positives: **58,700**



These results show that the model correctly identified approximately 46% of the real future empty-station events. When the model predicted that a station would become empty, approximately 64% of those predictions were correct.

The test performance was relatively consistent with the validation results:

Metric	Validation	Test
Accuracy	93.17%	94.05%
Precision	67.50%	63.66%
Recall	48.28%	46.23%

11. Error Analysis

The final model still produced both false positives and false negatives. A false positive occurs when the model predicts that a station will become empty, but it does not actually reach zero bikes within the next 60 minutes.

These cases may occur when a station has very few bikes and shows a recent downward trend, making it appear risky, but the remaining bikes are not removed quickly enough to reach zero within the prediction window.

A false negative occurs when the model predicts that a station will not become empty, but it actually does. These cases are particularly important because they represent events that the model fails to detect. Some may involve stations that still have several bikes at prediction time but experience a relatively rapid decline afterwards.

	False Positive	False Negative
bikes	1.95	4.45
occupancy_ratio	0.08	0.19
bike_change_15m	-0.49	-0.25
bike_change_30m	-1.04	-0.45

Average feature values for false positive and false negative predictions

False positives generally occurred when stations had fewer bikes and a stronger recent downward trend. False negatives occurred when stations still appeared relatively safe, with more available bikes and a higher occupancy ratio.

12. Limitations

The first limitation concerns data coverage. There was a clear gap around late September and early October 2018. More importantly, the number of unique stations began to decline after early January 2019, with a strong reduction during February and a severe collapse in March. Because of this, March was excluded from the main modelling evaluation.

February was retained as the test set because it still contained more than 1.7 million eligible observations and its positive rate of 7.42% remained substantially more comparable with the training and validation periods than the 1.8% observed in March.

A second limitation is that the dataset contains station status snapshots rather than individual trip records. The model can observe that the number of bicycles changes, but it cannot determine whether a change was caused by a user trip, redistribution operation, maintenance action, or another process.

A third limitation is the absence of external variables. Factors such as weather, public holidays, large events, public transport disruptions, and local activity patterns may affect bicycle demand.

13. Future Improvements

The first would be to obtain a more complete and consistent dataset with stable station coverage over time. This would allow more reliable evaluation across later months.

Additional external variables could also be incorporated, including:

- Weather conditions
- Rainfall
- Temperature
- Holidays
- Major events
- Public transport disruptions

Another possibility would be to predict probability rather than only a binary class and use different operational thresholds depending on the cost of false positives and false negatives.

Geographic distribution of nearby stations could be incorporated. A station may be affected not only by its own recent history but also by the availability and demand patterns of surrounding stations.

14. Conclusion

This project investigated whether Barcelona Bicing stations could be predicted to become empty within the following 60 minutes using current station conditions, recent availability history, temporal variables, and location.

Six predictive approaches were compared: Dummy Classifier, rule-based baseline, Logistic Regression, Decision Tree, Random Forest, and Neural Network.

The Random Forest achieved the highest validation accuracy and precision, while the rule based baseline achieved the highest recall. However, the Decision Tree offered the most suitable balance between precision and recall and was therefore selected as the final model.

On the unseen February test set, the Decision Tree achieved:

- **94.05% Accuracy**
- **63.66% Precision**
- **46.23% Recall**

The results show that short-term station emptying risk can be predicted to a meaningful extent using current availability and recent historical patterns. However, the model still misses a substantial percentage of real empty station events.

The project also revealed an important data quality issue: station coverage declined considerably during the later months, particularly in March. This had a strong effect on the target distribution and demonstrated why temporal data should be evaluated carefully rather than randomly split.

Overall, the project shows that a relatively interpretable model such as a Decision Tree can provide useful predictions for a real urban mobility problem.